

SOEC Digitization and Analysis Report

Digitization and Loss Analysis of a Solid Oxide Electrolysis Cell (SOEC) Polarization Curve

Author: Benyamin Mohammadzadeh Madiyeh **Affiliation:** Applied Chemistry, K.N. Toosi University of Technology

1. Introduction

Securing sustainable and efficient energy has long been one of humanity's most persistent challenges. Population growth, industrialization, and rising energy demand have driven extensive reliance on fossil fuels; despite their historical role in societal progress, their widespread consumption has caused significant environmental impacts, including air pollution and the intensification of global warming. Consequently, developing clean, efficient, and sustainable energy conversion technologies has become a central priority in scientific research.

As someone interested in chemistry and energy, I aim to contribute — on however modest a scale — to this global challenge through research in clean energy technologies. Solid oxide cells, which enable efficient conversion between chemical and electrical energy (as fuel cells) or the reverse process of hydrogen production (as electrolysis cells), have attracted considerable attention for their high efficiency and potential to reduce emissions. Advancing and optimizing these systems represents a meaningful step toward reducing dependence on fossil fuels and building a cleaner, more sustainable energy system.

Beyond its connection to solid oxide cell research, this project provided an opportunity to apply Python programming to a real scientific problem, as a self-directed exercise undertaken in parallel with formal coursework. A central goal was to become familiar with the process of analyzing literature-derived experimental data and using computational methods to extract meaningful results — specifically, digitizing and processing polarization-curve data and visualizing trends in cell performance. This experience is intended as a foundation for more extensive future work in energy-system modeling, data analysis, and optimization.

2. Methodology

2.1 Data Source

Voltage-current density data were digitized from Figure S5 (Ebbesen & Mogensen, 2009 dataset, 850°C, 50% H₂O) in the supplementary material of: Estrada, N.G.A. & Cervera, R.B.M. (2025). *Machine Learning-Based Predictive Modelling for Solid Oxide Electrolysis Cell's (SOEC) Electrochemical Performance*. Applied Sciences, 15.

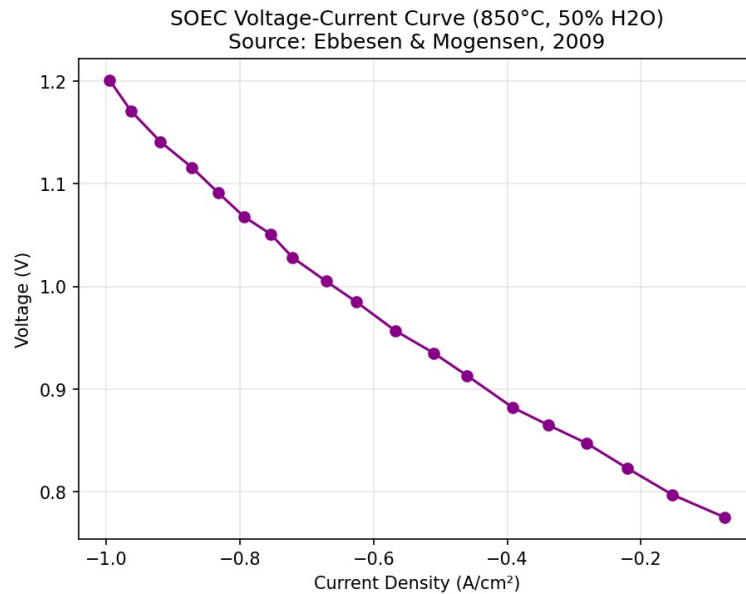
2.2 Digitization Procedure

Data points were extracted using a custom Python image-processing pipeline: pixel coordinates of marker centroids were identified via color thresholding and morphological filtering, then converted to data coordinates using axis tick-mark pixel positions as calibration references (equivalent in principle to manual digitization tools such as WebPlotDigitizer).

2.3 Analysis

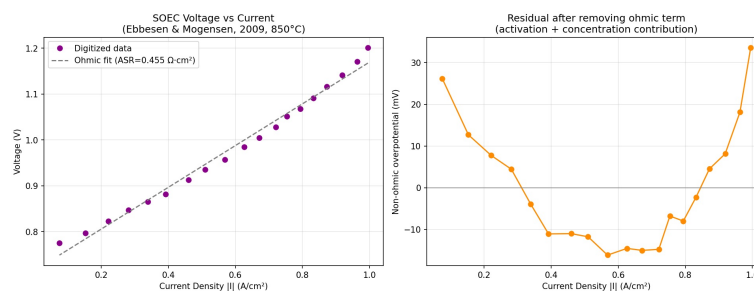
Voltage vs. current density data were fit to a linear (ohmic) model: $V = V_0 + \text{ASR} \times I$. The residual ($V - V_{\text{ohmic}}$) was examined as a function of current density to qualitatively separate the ohmic contribution from the combined activation and concentration overpotentials.

3. Results



Polarization curve

Figure 1. Digitized SOEC voltage vs. current density curve (850°C, 50% H₂O).



Loss separation

Figure 2. Left: data with linear (ohmic) fit. Right: residual (non-ohmic) overpotential vs. current density.

- Linear (ohmic) fit: $V = 0.715 + 0.455 \times I$ ($R^2 = 0.987$)
- Apparent area-specific resistance (ASR) $\approx 0.455 \Omega \cdot \text{cm}^2$
- Residual shows a U-shaped trend: positive at low I (activation-dominated) and positive/increasing at high I (concentration-dominated), near zero in the mid-range (ohmic-dominated)

3.1 Temperature Dependence (Extended Analysis)

To extend the analysis beyond a single operating condition, a second figure (Sun et al., 2013 dataset) was digitized, providing polarization curves at three temperatures (750°C, 800°C, 850°C, all at 50% H₂O) — 53 additional data points in total.

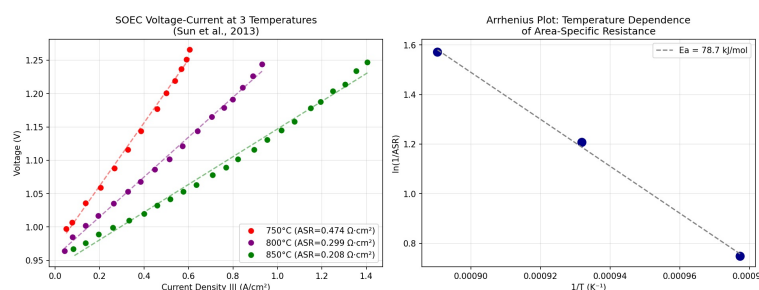


Figure 3. Left: polarization curves at three temperatures with linear (ohmic) fits. Right: Arrhenius plot of $\ln(1/ASR)$ vs. $1/T$.

Temperature	ASR ($\Omega\cdot\text{cm}^2$)
750°C	0.474
800°C	0.299
850°C	0.208

As expected physically, ASR decreases monotonically with increasing temperature, consistent with thermally activated ionic conduction in the solid oxide electrolyte. Fitting an Arrhenius relation ($\ln(1/ASR)$ vs. $1/T$) to the three data points yields an apparent activation energy of **$E_a \approx 78.7$ kJ/mol**, which is consistent with literature-reported activation energies for solid oxide electrolyte ionic conduction (typically 80–100 kJ/mol), providing an independent physical consistency check on the digitized data and analysis pipeline.

4. Discussion

The dominance of the ohmic contribution ($R^2 = 0.987$ for the linear fit alone) suggests that, for this particular SOEC operating condition, cell performance is primarily limited by the material resistance of the cell components rather than by reaction kinetics or mass transport — although the U-shaped residual indicates that activation and concentration effects remain non-negligible at the low- and high-current extremes, respectively.

The extended temperature-dependence analysis (Section 3.1) reinforces this picture: ASR decreases substantially with increasing temperature (0.474 $\rightarrow$ 0.208 $\Omega\cdot\text{cm}^2$ from 750°C to 850°C), and the extracted activation energy (≈ 78.7 kJ/mol) falls within the range typically reported for solid oxide electrolyte ionic conduction. This agreement between an independently derived quantity and literature values is a useful internal check — it suggests that the digitization and fitting pipeline produces physically meaningful, not merely numerically convenient, results.

From an engineering standpoint, reducing ohmic resistance could in principle be pursued through three routes: (i) decreasing electrolyte thickness, (ii) increasing operating temperature, or (iii) using an electrolyte material with higher intrinsic ionic conductivity. The first two routes involve significant trade-offs: a thinner electrolyte reduces mechanical durability and increases replacement costs (which is why modern designs typically use a thin electrolyte supported on a thicker anode layer, balancing low resistance against structural integrity), while higher operating temperatures are constrained by material degradation and sealing costs. Substituting the electrolyte material — for example, with scandia-stabilized zirconia (ScSZ) or gadolinium-doped ceria (GDC), both of which offer higher ionic conductivity than conventional yttria-stabilized zirconia (YSZ) — appears to be the most promising route, as it directly targets the dominant loss mechanism without the durability and cost trade-offs associated with electrolyte thinning. This is consistent with current research trends in solid oxide cell materials.

5. Limitations and Future Work

- This is a simplified two-term (ohmic vs. non-ohmic) decomposition rather than a full three-parameter (activation/ohmic/concentration) fit, since exchange current density and limiting current density were not available from the source figure alone.
- The dataset now covers two source curves (Ebbesen & Mogensen, 2009 and Sun et al., 2013) across four operating conditions (one at 850°C/50% H₂O, three spanning 750–850°C). The remaining 10 papers compiled in the same supplementary dataset could be digitized similarly for a broader comparative study across cell architectures and geometries.

References

1. Ebbesen, S.D. & Mogensen, M. (2009). Electrolysis of carbon dioxide in Solid Oxide Electrolysis Cells. *Journal of Power Sources*, 193(1), 349–358.
2. Estrada, N.G.A. & Cervera, R.B.M. (2025). Machine Learning-Based Predictive Modelling for Solid Oxide Electrolysis Cell's (SOEC) Electrochemical Performance. *Applied Sciences*, 15(17), 9388.